

Phonological change without phonetic change? The resyllabification of yod in American English

Aidan Malanoski¹, Bill Haddican^{1,2}, Kyle Gorman¹, Eniya Exil², Cynthia Gan², Jack Lacey², Jack Lynch², Elissa Patin², Donna Shair², Samuel Sokol², Ioannis Stathopoulos²

¹CUNY Graduate Center

²Queens College, CUNY

Introduction

Introduction

- We investigate the syllabification of yod in /\$C₁jV/ sequences in American English (AmE)
- We find that yod is increasingly being parsed in the nucleus rather than the onset

The phonotactics of /\$C₁jV/

- Historically, /\$C₁jV/ sequences come from the /iw/ diphthong (Harris 1994; Labov, Ash & Boberg 2006 [henceforth ANAE])
- Subsequent phonological changes have made yod phonotactically ambiguous in this environment (Harris 1994; Barlow 2001)
- There is interspeaker variability in American English (Barlow 1996, 2001)

The phonotactics of /\$C₁jV/

Onset

- Does not follow:
 - consonant clusters (e.g., *true*, *blue*) [most dialects; Harris 1994]
 - liquids (e.g., *rude*, *lewd*) [many dialects; ANAE]
 - coronals (e.g., *news*) [North American dialects; ANAE]

Nucleus

- Only occurs with GOOSE and CURE*
- Can follow nasals and /v/ (e.g., *mew*, *view*)

*In some dialects, [j] occurs between a velar stop and a following low vowel (e.g., *garden*; Kurath & McDavid 1961). Additionally, some speakers allow otherwise illicit /CjV/ sequences in loan words (e.g., *gyoza*).

Pilot study (Malanoski et al. 2025)

- Found that younger AmE speakers are less likely to say that /Cjuw/-/uw/ pairs rhyme
- **Resyllabification hypothesis:** Younger speakers are more likely to parse /j/ in the nucleus; difference in nucleus blocks rhyme
- **Allophony hypothesis:** Younger speakers are more likely to have a distinct allophone of /uw/ after /j/; allophonic difference blocks rhyme

Outline

1. Introduction
2. Methods
3. Results
4. Conclusion

Methods

Overview

- Survey consisted of:
 - Rhyme judgment task
 - Language game task
 - Demographic questionnaire
- Administered on Google Forms in spring 2025
 - Question order randomized for each participant
- Convenience sample recruited via word-of-mouth and snowballing

Rhyme judgments

- Presented in writing
- Binary forced choice

goose/loose *

☐ yes

☐ no

☐ I don't know how to pronounce one/both of the words

Rhyme stimuli

- Eight classes of test items (/K/ = non-coronal consonant; /T/ = coronal consonant):
 - 5 /CwV/-/CV/ (e.g., *quick-pick*)
 - 7 /Cjuw/-/Kuw/ (e.g., *cute-boot*)
 - 4 /Cjuwl/-/Kuwl/ (e.g., *fuel-pool*)
 - 7 /Cjuw/-/Tuw/ (e.g., *fume-doom*)
 - 5 /Tuw/-/Kuw/ (e.g., *soup-goop*)
 - 3 /juw/-/Kuw/ (e.g., *you-coo*)
 - 3 /juw/-/Tuw/ (e.g., *youth-tooth*)
 - 5 /Cjuw/-/Cjuw/ (e.g., *few-mew*)
- Two classes of catch items:
 - 5 expected positives (e.g., *see-flea*)
 - 10 expected negatives (e.g., *peep-dip*)

“Cow Latin”

- Pig Latin, but ending in [uw] instead of [ej]: *kit* [kɪt] → [ɪtkuw]
 - Designed to avoid the confound that /j/ does not occur in the environment $\$C_1_ej$ (cf. Barlow 2001)
- Intended to test whether yod is parsed with the onset (moved: *cute* → “oot-kyoo”) or the nucleus (unmoved: *cute* → “yoot-koo”)
- Task preceded by training session
 - Taught to move entire onset (e.g., *true* → *oo-troo*)

“Cow Latin” stimuli

- Two classes of test items:
 - 14x /Cjuw/ (e.g., *cube*)
 - 6x /juw/ (e.g., *you*)
- Four classes of catch items:
 - 5x /CrV/ (e.g., *price*)
 - 5x /ClV/ (e.g., *plate*)
 - 5x digraph catch items (e.g., *photo*)
 - 5x filler (e.g., *kit*)

“Cow Latin”

few *

- ☐ fyoo-oo
- ☐ oo-fyoo
- ☐ yoo-foo
- ☐ I don't know this word/I don't know how to pronounce this word

Demographic questions

- Data elicited:
 - Birth year (open response)
 - Race/ethnicity (multiple choice with open response option)
 - Gender (multiple choice with open response option)
 - Education (multiple choice with open response option)
 - Childhood home (zip code; open response)
 - Current home (zip code; open response)
 - Childhood home language (multiple choice with open response option)
 - Other (open response)

Sample

- 82 respondents; 50 retained
- Excluded participants who:
 - answered “no” for 1+ expected positive rhymes
 - answered “yes” for 1+ expected negative rhymes
 - self-reported that they did not understand the Cow Latin task
 - gave the desired response for less than 90% of the cluster and digraph Cow Latin items

Sample

- Self-reported native English speakers
- 41 white; 2 Asian; 3 Hispanic/Latine; 3 mixed/multi; 1 “Jewish”
- 28 women; 19 men; 3 non-binary
- Speakers mostly from Eastern US

What would support the resyllabification hypothesis?

- Correlation between /Cjuw/-/uw/ rhyme rate and Cow Latin responses (higher rhyme rate ~ more [uw-Cjuw] responses)
- Different patterning of /Cjuw/ vs. /juw/

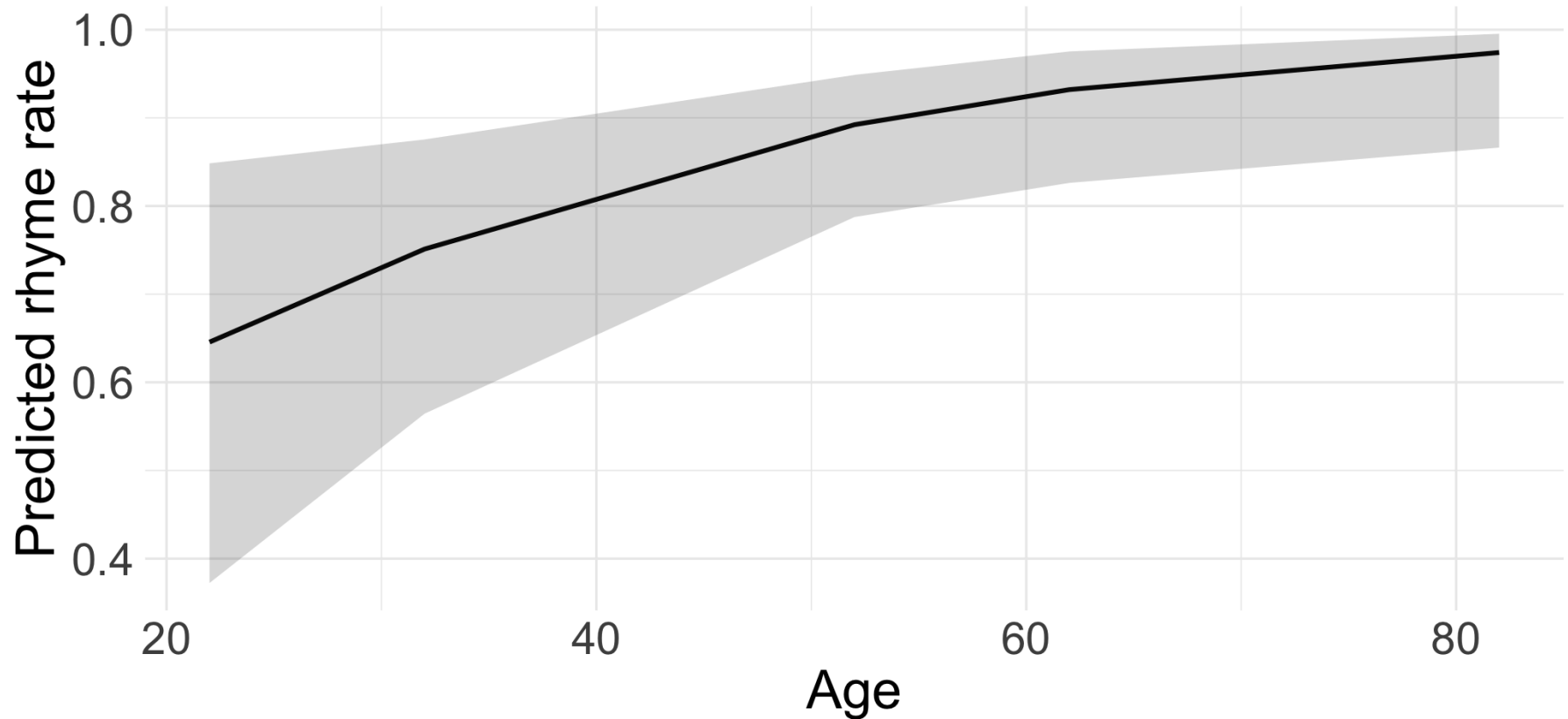
What would support the allophony hypothesis?

- Non-ceiling /Tuw/-/Kuw/ rhyme rate
- Different rhyme rate for /Cjuw/-/Tuw/ vs. /Cjuw/-/Kuw/ pairs
- No correlation between /Cjuw/-/uw/ rhyme rate and Cow Latin responses
- Parallel patterning of /Cjuw/ vs. /juw/ in rhyme judgments

Results

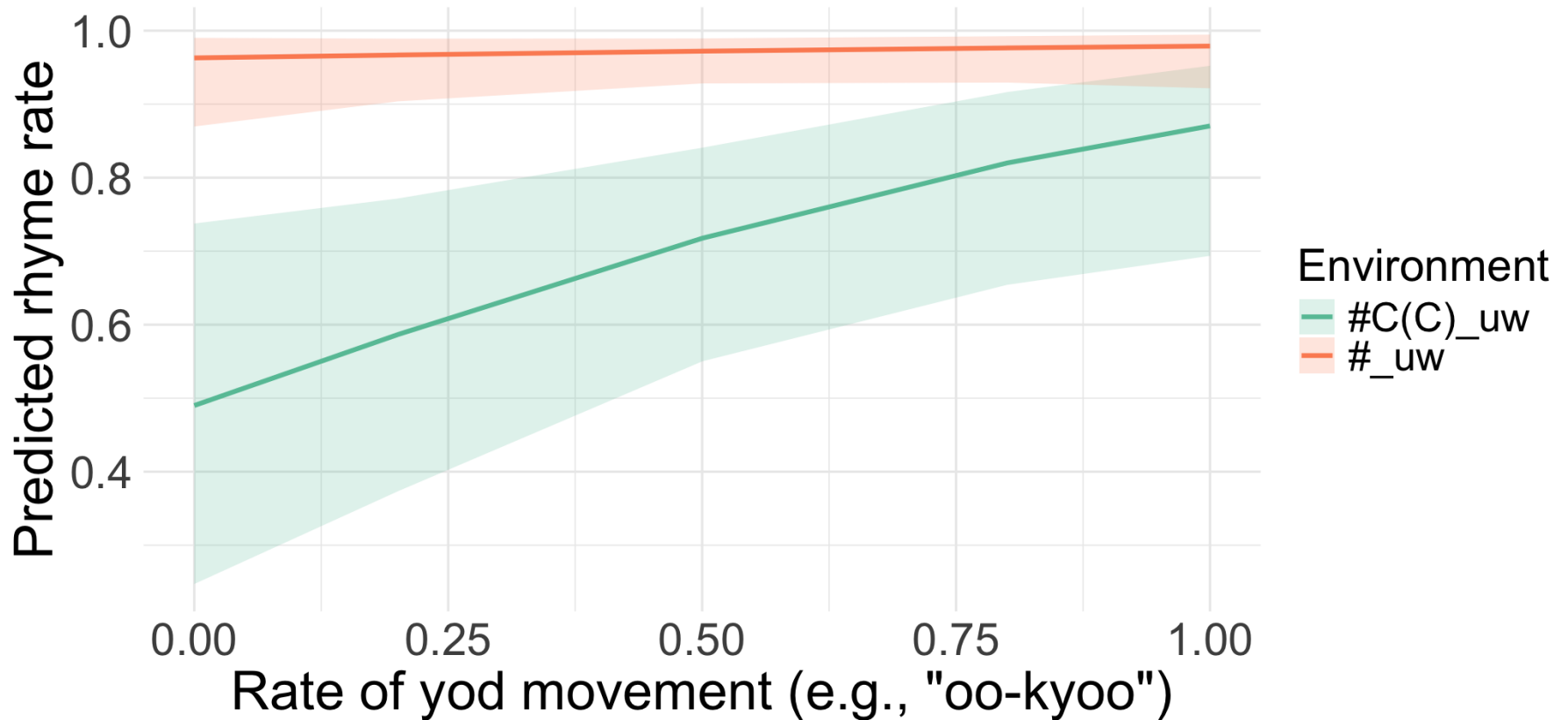
Younger participants are less likely to accept /Cjuw/-/uw/ rhymes

Effect of age on rhyme rate



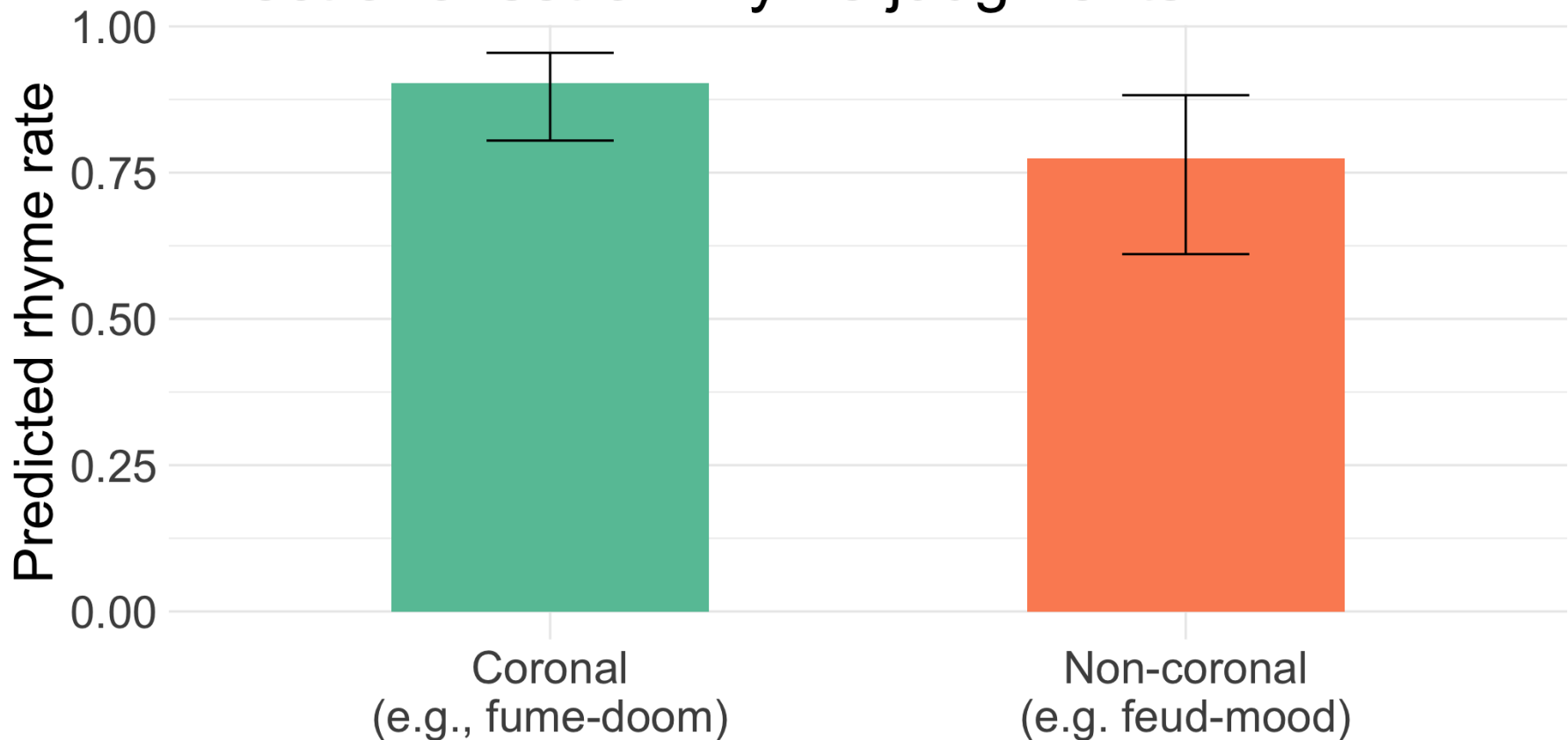
Cow Latin responses predict rhyme judgments

Effect of Cow Latin response on rhyme judgments

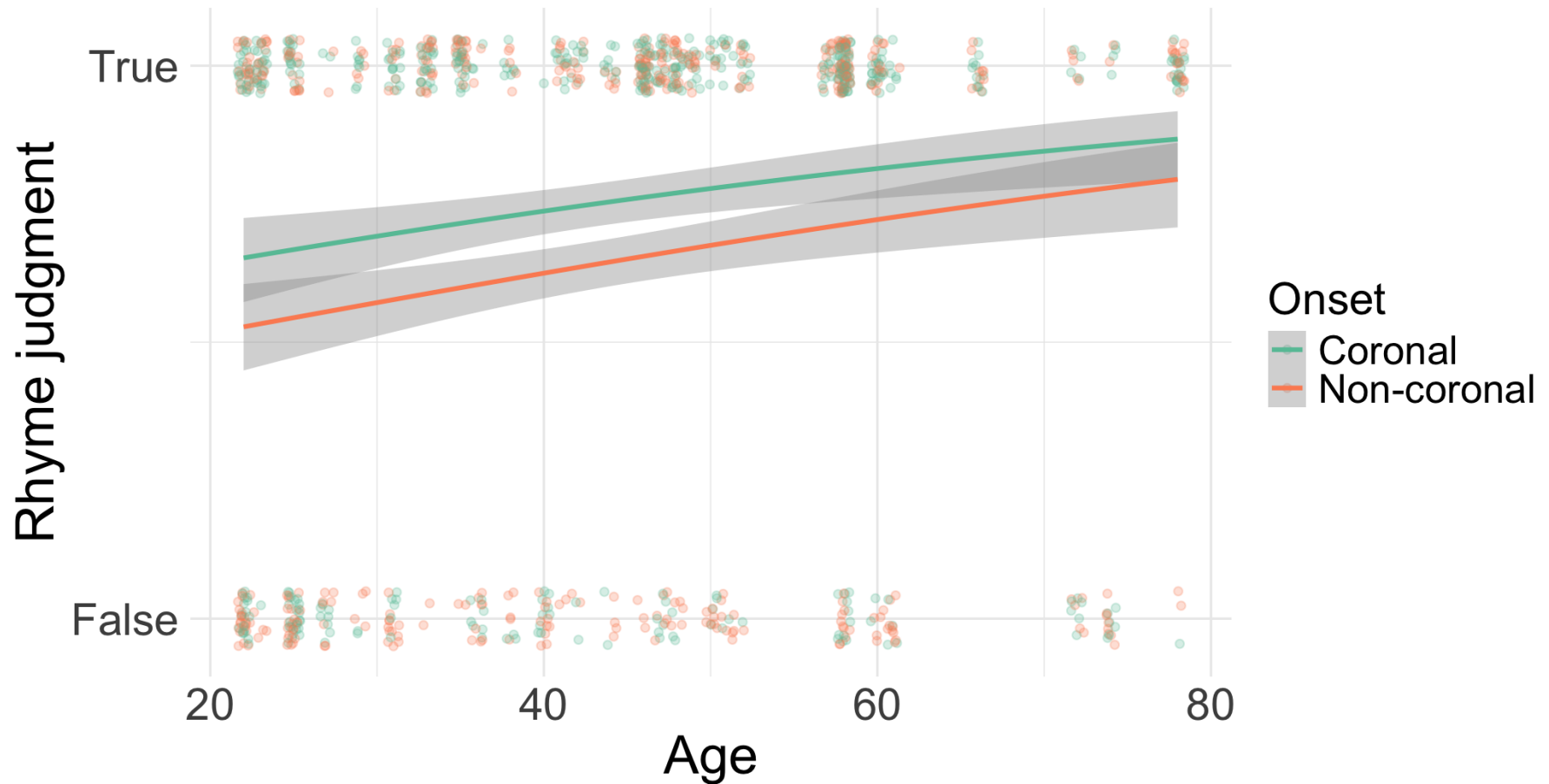


/Cjuw/ and /juw/ items are more likely to rhyme with post-coronal than non-post-coronal GOOSE...

Effect of onset on rhyme judgments

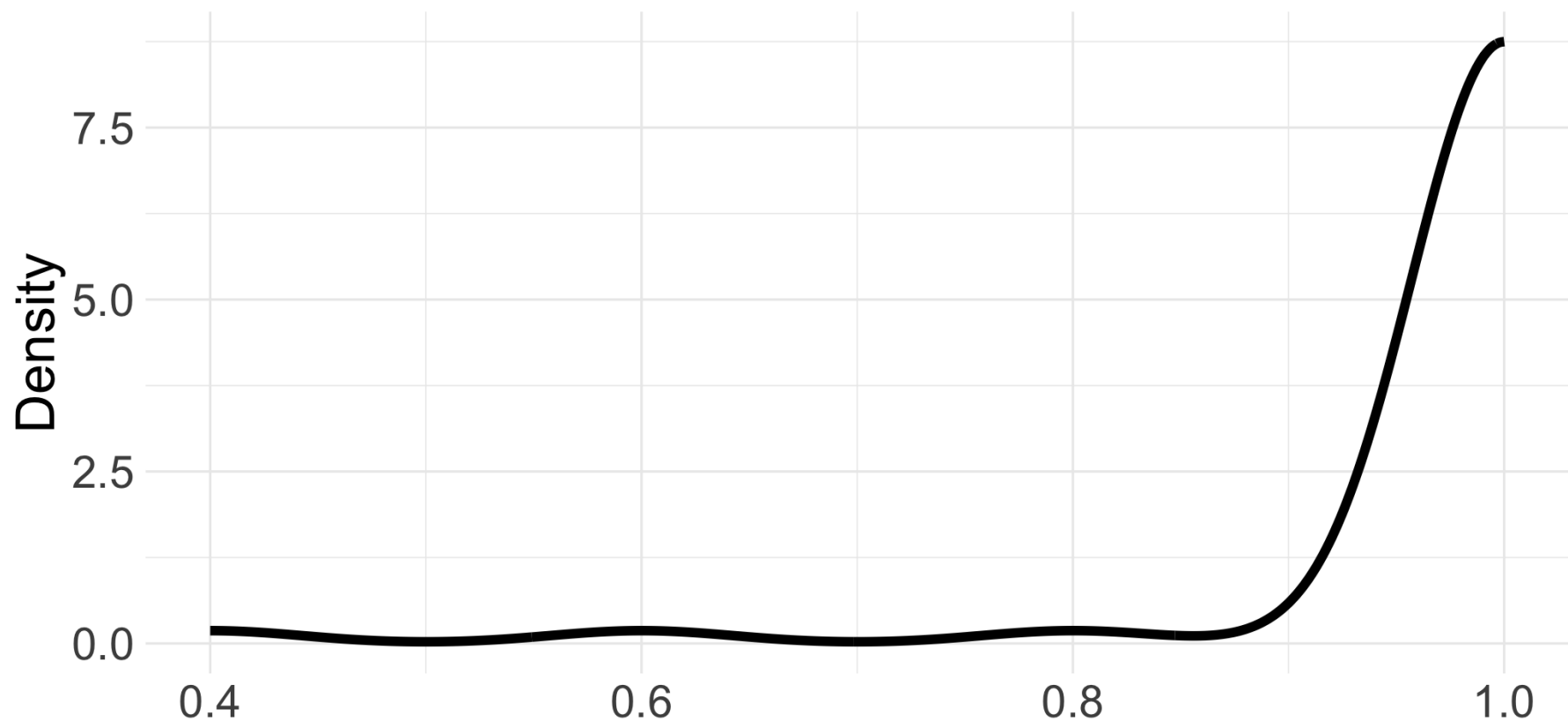


But there is no interaction with age



The /Tuw/-/Kuw/ rhyme rate is at ceiling

Rate of positive /Tuw/-/Kuw/ rhyme judgments



What would support the resyllabification hypothesis?

- ✓ Correlation between /Cjuw/-/uw/ rhyme rate and Cow Latin responses (higher rhyme rate ~ more [uw-Cjuw] responses)
- ✓ Different patterning of /Cjuw/ vs. /juw/

What would support the allophony hypothesis?

- ✗ Non-ceiling /Tuw/-/Kuw/ rhyme rate
- ✓ Different rhyme rate for /Cjuw/-/Tuw/ vs. /Cjuw/-/Kuw/ pairs
 - Diachronically stable → does not account for change
- ✗ No correlation between /Cjuw/-/uw/ rhyme rate and Cow Latin responses
- ✗ Parallel patterning of /Cjuw/ and /juw/ in rhyme judgments

Conclusion

Main findings

1. Younger speakers are more likely to reject /Cjuw/-/uw/ rhymes
2. This likely reflects a change in parsing (resyllabification hypothesis) rather than a change in realization (allophony hypothesis)

The actuation problem

- The phonotactic evidence is relatively stable
 - If anything, an onset parse would be favored even more strongly among younger speakers
- Phonotactic ambiguity alone cannot be responsible for this change
- There must be something else at play

The actuation problem

- What else could be at play?
 - Phonological reorganization
 - But what would trigger it?
 - Phonetic change
 - Post-coronal fronting?
 - Gestural retiming? (Margaret Renwick, p.c.)

Acknowledgments

- Thank you to:
 - Our participants
 - Keiko Bridwell
 - Austin Johnson
 - Peggy Renwick
 - ADS audience

References

- Barlow, Jessica A. 1996. The development of on glides in American English. *Proceedings of the 20th annual Boston University Conference on Language Development*, 40–51. Somerville: Cascadilla Proceedings Press.
- Barlow, Jessica A. 2001. Individual differences in the production of initial consonant sequences in Pig Latin. *Lingua* 111. 667–696.
- Harris, John. 1994. *English sound structure*. Oxford: Blackwell.
- Kurath, Hans & Raven I. McDavid, Jr. 1961. *The pronunciation of English in the Atlantic states*. Ann Arbor: University of Michigan Press.
- Labov, William, Sharon Ash & Charles Boberg. 2006. *The atlas of North American English: Phonetics, phonology and sound change*. Berlin: Mouton De Gruyter.
- Malanoski, Aidan, Bill Haddican, Kyle Gorman, Cynthia Gan, Jack Lacey, Jack Lynch, Andrew Shillingford, Samuel H. Sokol & Kujegé Thiam. January 11, 2025. A whole new /iw/: The changing syllabification of yod [Poster presentation]. Annual Meeting of the American Dialect Society, Philadelphia, PA.

Appendix A: Cow Latin instructions

For this part of the survey, you will be asked to turn English words into Cow Latin, which is similar to Pig Latin. The main difference between Cow Latin and Pig Latin is that instead of ending in "ay", Cow Latin words end in "oo." In other words, you take the beginning of the word and move it to the end, then add "oo." For example, *yank* in Cow Latin is *ank-yoo*, which rhymes with *thank you*. Here's some more examples of words converted to Cow Latin:

- cat -> at-koo
- fill -> ill-foo
- cheese -> eez-choo
- gnome -> ome-noo

Make sure you move the **whole beginning** of each word:

- crust -> ust-kroo
- bleak -> eek-bloo
- quack -> ack-kwoo

If the word begins with a vowel sound, then you just add "oo":

- eye -> eye-oo
- herb -> erb-oo

We will present you with various options for how to turn a word into Cow Latin, and we want you to choose the one you think is correct. We will spell the options in a way that reflects their pronunciation.

We'll begin with some practice words.

Appendix B: Model predicting /Cjuw/- /Cuw/ rhyme judgments

	Estimate	Std. error	z value	Pr(> z)
Intercept	−0.674	0.793	−0.849	0.396
Onset: Coronal	−1.000	0.362	−2.759	0.006
Yod environment: #_uw	3.297	0.549	6.009	<0.001
Rate of yod movement	1.945	0.811	2.397	0.017
Age	0.050	0.021	2.400	0.016
Yod-initial item * Rate of yod movement	−1.358	0.646	−2.102	0.036

```
glmer(Rhyme ~ 1 + Onset + Age + Yod environment*Rate of yod
movement + (1 | Participant) + (1 | Pair)). N=50. Obs=990.
```